Robust Control Implementation for Autonomous Vehicles in Urban Scenarios: Signal-Based Navigation and $\mathcal{H}_2$ vs $\mathcal{H}_\infty$ Performance Analysis

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Abstract

This report presents a simulation study of an autonomous vehicle navigating a typical urban environment using MATLAB's Automated Driving Toolbox. The vehicle detects lane markings and traffic signals, and autonomously executes maneuvers such as steering, stopping, and speed regulation. Two robust control strategies, $\mathcal{H}_2$ and $\mathcal{H}_\infty$, are implemented and compared in terms of trajectory tracking, control effort, and overall performance under uncertainties. The results highlight the strengths and limitations of each control approach, providing insights into the selection of robust controllers for urban autonomous driving applications.

Keywords: Autonomous Vehicles, Robust Control, LPV Systems, $\mathcal{H}_2/\mathcal{H}_\infty$ Control, Urban Maneuvering, Trajectory Tracking.

1. Introduction

Autonomous vehicles are rapidly transforming urban transportation, promising improvements in safety, traffic efficiency, and driving comfort. Ensuring reliable and safe operation in complex urban scenarios requires effective control strategies capable of handling model uncertainties and external disturbances. This project simulates a typical urban driving scenario using MATLAB's Automated Driving Toolbox, assuming that the vehicle can detect signals embedded in the road surface. Guided by a custom-designed trajectory, the vehicle autonomously performs maneuvers such as steering, stopping, and maintaining the desired speed. Two robust controllers, $\mathcal{H}_2$ and $\mathcal{H}_\infty$, are implemented to govern the vehicle's behavior. The study focuses on evaluating and comparing their performance in terms of trajectory tracking, control effort, and robustness, providing insights into the most suitable control approach for autonomous urban driving.

2. Model Description

The vehicle dynamics in this project are modeled using a simplified bicycle model, which captures the essential behavior of a car in planar motion while remaining computationally efficient for simulation.
The state variables of the system are the vehicle's position (x, y) and orientation (yaw angle) ψ, and the inputs are the front and rear steering angles δ_f and δ_r. The nonlinear kinematic equations governing the system are:

$$\begin{cases} \dot{x} = V_x \cos(\psi + \beta) \\ \dot{y} = V_x \sin(\psi + \beta) \\ \dot{\psi} = \frac{V_x \cos(\beta)}{l_f + l_r} \left(\tan(\delta_f) - \tan(\delta_r)\right) \end{cases} \tag{1}$$

where

- $\boldsymbol{V_x}$ is the longitudinal velocity of the vehicle
- $\boldsymbol{\beta}$ is the vehicle sideslip angle
- $\boldsymbol{l_f}$ and $\boldsymbol{l_r}$ are the distances from the center of gravity to the front and rear axles, respectively
- $\boldsymbol{\delta_f}$ and $\boldsymbol{\delta_r}$ are the front and rear steering angles.

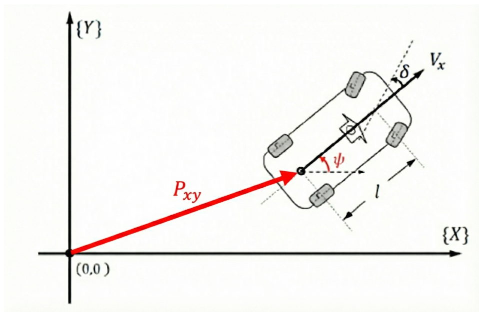

Figure 1: Kinematic Bicycle Model

To simplify the vehicle model, the sideslip angle is assumed to be negligible ($\beta = 0$), and rear-wheel steering is not considered ($\delta_r = 0$). Under these assumptions, the vehicle is described by a front-steering kinematic bicycle model.
An external disturbance is also considered to account for unmodeled dynamics and environmental effects.
Moreover, the non-linear model will be:

$$\begin{cases} \dot{x} = V_x \cos(\psi) \\ \dot{y} = V_x \sin(\psi) \\ \dot{\psi} = \frac{V_x}{l_f + l_r} \left(\tan(\delta_f) + a\,\omega(t)\right) \end{cases} \tag{2}$$

with $a = 0.1$.

2.1. State Space Representation and Linear Model

Consider small-angle, such that $\sin(\psi) \approx \psi$ and $\tan(\delta_f) \approx \delta_f$, then

$$\mathrm{x} = \begin{bmatrix} y \\ \psi \end{bmatrix} \quad u = \delta_f$$

The linearized state-space model is given by:

$$\dot{\mathrm{x}} = A\mathrm{x} + B_1 u + B_2 \omega$$

where, defining $L = l_r + l_f = 2.8$ m,

$$A = \begin{bmatrix} 0 & V_x \\ 0 & 0 \end{bmatrix}, \qquad B_1 = \begin{bmatrix} 0 \\ \frac{V_x}{L} \end{bmatrix} \qquad B_2 = a\, B_1 \tag{3}$$

2.2. Linear Parametric-Varying (LPV) Model

In urban environments, variations in longitudinal velocity significantly affect vehicle dynamics. To capture this dependency and ensure robust controller performance over a range of operating conditions, the vehicle model is expressed in a Linear Parameter-Varying (LPV) form, with the longitudinal velocity V_x used as the scheduling parameter.

$$\begin{cases} \dot{\mathrm{x}}(t) = A(p(t))\mathrm{x}(t) + B_1(p(t))u(t) + B_2(p(t))\omega(t) \\ \mathrm{y}(t) = C(p(t))\mathrm{x}(t) \end{cases} \tag{4}$$

where

$$p(t) = \begin{bmatrix} p_1(t) \\ \vdots \\ p_l(t) \end{bmatrix}, \qquad 0 \le p_i(t) \le 1, \qquad \sum_{i=1}^{l} p_i(t) = 1$$

$$A(p(t)) = \sum_{i=1}^{l} P_i(t)A_i, \qquad B_1(p(t)) = \sum_{i=1}^{l} p_i(t)B_{1,i}$$

$$B_2(p(t)) = \sum_{i=1}^{l} p_i(t)B_{2,i}, \qquad C(p(t)) = \sum_{i=1}^{l} p_i(t)C_i$$

In the simulation, the vehicle longitudinal velocity varies from 0 to 5 m/s to reproduce realistic urban driving conditions.
To improve LPV approximation, this range is divided into six sub-intervals

$$\zeta_1 : [0,\, 0.8] \text{ m/s} \qquad \zeta_2 : [0.8,\, 1.6] \text{ m/s} \qquad \zeta_3 : [1.6,\, 2.4] \text{ m/s}$$
$$\zeta_4 : [2.4,\, 3.2] \text{ m/s} \qquad \zeta_5 : [3.2,\, 4.1] \text{ m/s} \qquad \zeta_6 : [4.1,\, 5] \text{ m/s}$$

To weight the LPV models within each velocity interval, a normalized auxiliary function $\chi^j(t)$ maps the current speed $V_x(t)$; indicating the j-th interval $\left[\underline{V}_x^j,\, \overline{V}_x^j\right]$:

$$\chi^j(t) = \frac{V_x(t) - \underline{V}_x^j}{\overline{V}_x^j - \underline{V}_x^j} \qquad 0 \le \chi^j(t) \le 1 \tag{5}$$

Two local weights are then defined for interpolation between interval vertices:

$$\begin{cases} p_1^j = \chi^j \\ p_2^j = 1 - p_1^j \end{cases} \tag{6}$$

with $j = 1, \cdots, 6$
The global scheduling vector p is finally computed by combining these local weights with the sigmoid functions $\sigma_i(V_x)$ which select the active interval:

$$p = \sigma_{[i/6]}(V_x)p_i^{[i/6]}, \qquad i = 1, 2 \tag{7}$$

where

$$\sigma_j(V_x) = \left(1 + e^{-b(V_x - \underline{V}_x^j)}\right)^{-1} - \left(1 + e^{-b(V_x - \overline{V}_x^j)}\right)^{-1} \tag{8}$$

The use of sigmoid functions ensure smooth transitions between intervals.
Finally, the global LPV controller is then obtained as a a weighted combination of local controllers:

$$K(p) = \sum_{j=1}^{6} p_j(t)Y_jX^{-1} \tag{9}$$

Considering the parameter $b = 10$

$$\sigma_j(V_x) = \begin{cases} \sigma_1(V_x) = 1 - (1 + e^{-10(V_x - 0.8)})^{-1} \\ \sigma_2(V_x) = (1 + e^{-10(V_x - 0.8)})^{-1} - (1 + e^{-10(V_x - 1.6)})^{-1} \\ \sigma_3(V_x) = (1 + e^{-10(V_x - 1.6)})^{-1} - (1 + e^{-10(V_x - 2.4)})^{-1} \\ \sigma_4(V_x) = (1 + e^{-10(V_x - 2.4)})^{-1} - (1 + e^{-10(V_x - 3.2)})^{-1} \\ \sigma_5(V_x) = (1 + e^{-10(V_x - 3.2)})^{-1} - (1 + e^{-10(V_x - 4.1)})^{-1} \\ \sigma_6(V_x) = (1 + e^{-10(V_x - 4.1)})^{-1} - (1 + e^{-10(V_x - 5)})^{-1} \end{cases}$$

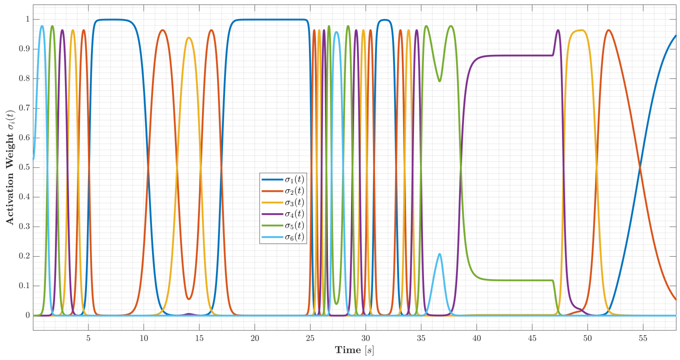

Figure 2: Sigmoid weighting functions

3. Simulation Scenario Description

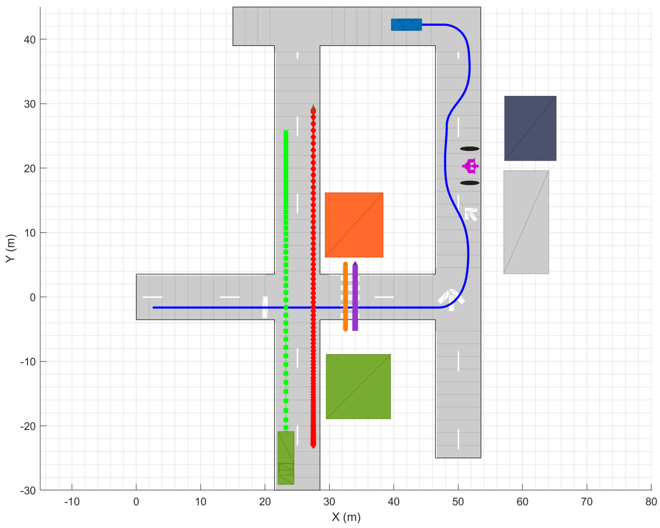

Figure 3: Simulation of the Urban Scenario

The simulated urban scenario consists of a sequence of maneuvers designed to test the autonomous vehicle's perception and control capabilities. In the figure, the autonomous vehicle is represented in blue.

The vehicle first approaches an intersection and detects the stop line, coming to a complete stop to wait for the crossing traffic (green truck and red motorcycle). Once the intersection is clear, the vehicle resumes motion.

Along its path, the vehicle encounters pedestrian crossings. It begins to slow down upon detecting the markings and stops when pedestrians (orange and purple) are present, allowing them to safely cross before continuing.

Next, the vehicle reaches an intersection with a mandatory left turn, indicated by a road marking arrow. After completing the turn, the vehicle encounters an obstacle on the lane. By detecting the obstacle and following the road marking indicating the bypass, the vehicle navigates around it and continues safely.

Finally, the vehicle comes to a stop immediately after the last curve, completing the scenario.

The complete path followed by the autonomous vehicle is highlighted in blue in Figure (3).

3.1. Speed Profile

The vehicle longitudinal speed V_x is modeled as **first-order low-pass filter** applied to the reference speed v_{ego}, which is provided by the Automated Driving Toolbox, to capture mechanical inertia and actuation delays. The relationship between the reference and the actual speed is described by:

$$\tau \dot{V}_x + V_x = v_{ego}(t)$$

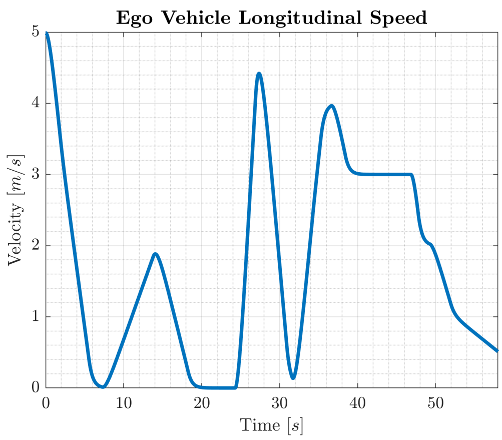

Figure 4: Speed of the Autonomous Vehicle

This filter transforms the discontinuous reference commands into a smooth and continuous speed profile.
The low-pass dynamics prevent instantaneous, unbounded accelerations and ensure an asymptotic, stable response.

4. Controller Design

The adopted control strategy is the **tracking problem**.
The main objective of the proposed control strategy is to ensure accurate tracking of a desired yaw reference by minimizing the corresponding tracking error. By regulating the yaw dynamics, the vehicle is able to follow the assigned trajectory in a stable and robust manner.
To achieve this goal, a state-feedback controller with augmented yaw error dynamics is adopted. This formulation allows the controller to explicitly minimize the yaw tracking error and its derivatives, improving both transient response and steady-state performance.

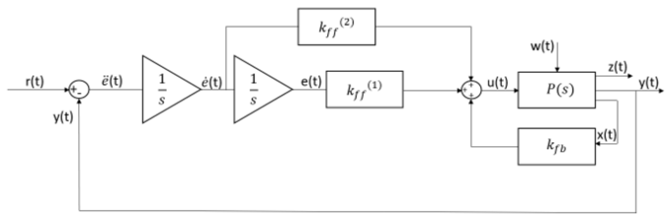

Figure 5: Control Scheme

Defining $\ddot{e} = r - \mathrm{y} = r - C\mathrm{x}$, the new *augmented state*, is

$$\mathrm{x}_a = \begin{bmatrix} \dot{e} & e & x \end{bmatrix}^T$$

and

$$u_a = u = \delta_f, \quad \omega_a = \begin{bmatrix} \omega & r \end{bmatrix}^T$$

the state space representation in the augmented scenario will be:

$$\begin{cases} \dot{\mathrm{x}}_a(t) = A_a \mathrm{x}_a + B_{1,a} u_a(t) + B_{2,a} \omega_a(t) \\ z(t) = C_{z,a} \mathrm{x}_a(t) + D_{zu,a} u_a(t) + D_{z\omega,a} \omega_a(t) \end{cases} \tag{10}$$

where

$$A_a = \begin{bmatrix} 0 & 0 & -C \\ 0 & 0 & 0 \\ 0 & 0 & A \end{bmatrix} \quad \text{with } C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$B_{1,a} = \begin{bmatrix} 0 & 0 & B \end{bmatrix}^T$$

$$B_{2,a} = \begin{bmatrix} 0 & 0 & B_2 \\ 1 & 0 & 0 \end{bmatrix}^T$$

$$C_{z,a} = \begin{bmatrix} 0 & 0 & Cz \end{bmatrix} \quad \text{with } C_z = \begin{bmatrix} 0 & \rho \end{bmatrix} \text{ and } \rho = 0.001$$

$$D_{zu,a} = D_{zu} = \sqrt{\rho}$$

$$D_{z\omega,a} = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

Finally, as shown in Figure (5), the control input is:

$$u = k_{ff}^{(2)} \dot{e} + k_{ff}^{(2)} e + K_{fb} \mathrm{x} \tag{11}$$

It is composed of a feedback term acting on the physical vehicle states and feedforward terms based on the yaw tracking error and its derivative.

5. Control Synthesis

To ensure stability and performance under model uncertainties, disturbances, and varying longitudinal speed, both $\mathcal{H}_2$ and $\mathcal{H}_\infty$ control strategies are adopted.
The controllers are designed within the LPV framework using the augmented state representation.

5.1. Disturbance Description

As shown in the Figure (6), the disturbance is modeled as a stochastic process with zero mean and variance equal to 10^{-2}, representing random external influences on the vehicle dynamics.

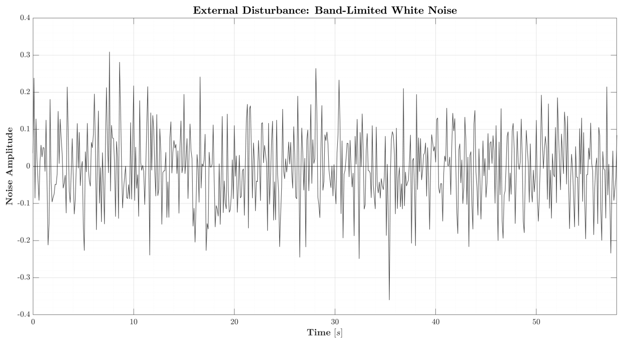

Figure 6: Realization of the disturbance

5.2. $\mathcal{H}_2$ Robust Optimal Control

The $\mathcal{H}_2$ control framework aims at minimizing the energy of the closed-loop system response to disturbances; formally

$$\|y\|_2 \leq \|G\|_2 \|\omega\|_2$$

where ω is the disturbance input, y is the performance output, and $\|G\|_2$ is the $\mathcal{H}_2$ norm of the system.
The $\mathcal{H}_2$ robust control problem is solved using *LMI* formulation:

$$\begin{aligned}
&[X^* \quad Y^*] = \arg \min_{X,Y,Q} \ \mathrm{tr}\{Q\} \\
&\text{s.t.} \quad \begin{bmatrix} (A_{a_i}X + B_{1,a_i}Y) + (A_{a_i}X + B_{1,a_i}Y)^T & B_{2,a_i} \\ B_{2,a_i}^T & -I_{nw} \end{bmatrix} < 0, \\
&\begin{bmatrix} Q & (C_{z,a}X + D_{zu,a}Y) \\ (C_{z,a}X + D_{zu,a}Y)^T & X \end{bmatrix} < 0, \\
&X = X^T > 0, \quad Q = Q^T > 0, \quad i = 1, \ldots, l.
\end{aligned}$$

The optimal controller gain is given by:

$$K_2^* = Y^* X^{*-1} = \begin{bmatrix} k_{ff,2}^{(2)} & k_{ff,2}^{(1)} & k_{fb,2} \end{bmatrix}$$

The following figures show the simulation results for the $\mathcal{H}_2$ controller.

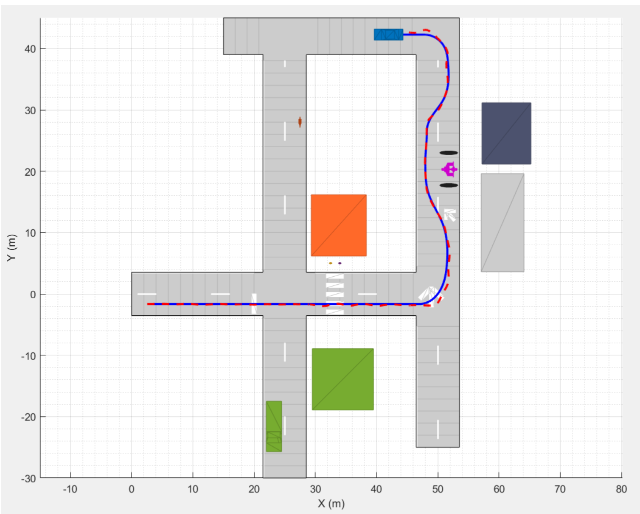

Figure 7: Vehicle trajectory with $\mathcal{H}_2$ controller

The oscillations in the signal in the Figure (8) indicate that the controller is responding to rapid disturbances and attempting to correct the tracking error very quickly.

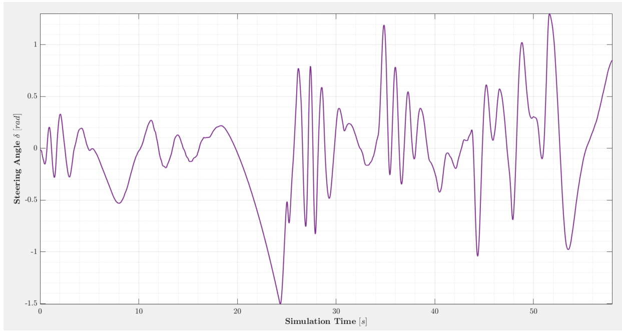

Figure 8: Front Steering angle with $\mathcal{H}_2$ controller

The system exhibits good tracking of the reference, confirming the effectiveness of the integral action in eliminating the steady-state error.

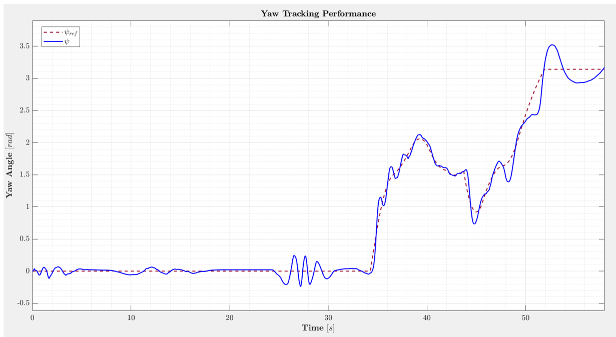

Figure 9: Yaw Angle Tracking with $\mathcal{H}_2$ controller

5.3. $\mathcal{H}_\infty$ Robust Optimal Control

The $\mathcal{H}_\infty$ control approach aims to minimize the worst-case effect of disturbances on the system; formally

$$\|y\|_2 \leq \|G\|_\infty \|u\|_2$$

where u and y denote the input and output signals, respectively.

The $\mathcal{H}_\infty$ robust control problem is solved using *LMI* formulation:

$$\begin{aligned}
&[X^* \quad Y^*] = \arg \min_{X,Y,\gamma} \ \gamma \\
&\text{s.t.} \quad \begin{bmatrix} (A_{a_i}X + B_{1,a_i}Y) + (A_{a_i}X + B_{1,a_i}Y)^T & B_{2,a_i} & (C_{z,a}X + D_{zu,a}Y)^T \\ B_{2,a_i}^T & -\gamma I_{nw} & D_{z\omega,a}^T \\ (C_{z,a}X + D_{zu,a}Y) & D_{z\omega,a} & -\gamma I_{nz} \end{bmatrix} < 0, \\
&X = X^T > 0, \quad i = 1, \ldots, l.
\end{aligned}$$

The optimal controller gain is given by:

$$K_\infty^* = Y^* X^{*-1} = \begin{bmatrix} k_{ff,\infty}^{(2)} & k_{ff,\infty}^{(1)} & k_{fb,\infty} \end{bmatrix}$$

The following figures show the simulation results for the $\mathcal{H}_\infty$ controller.

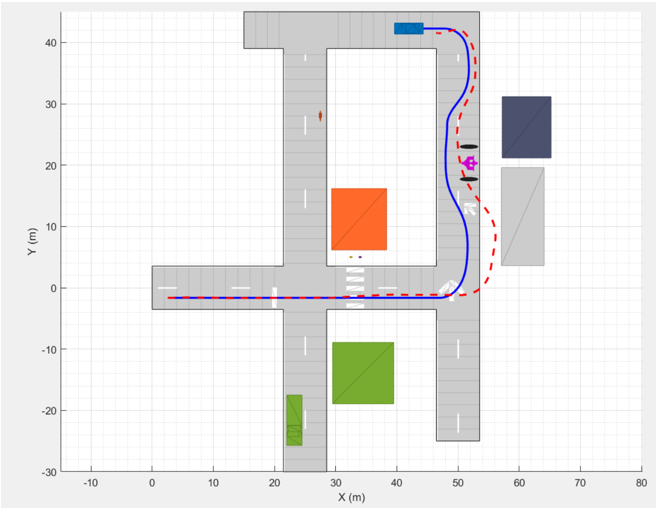

Figure 10: Vehicle trajectory with $\mathcal{H}_\infty$ controller

It can be visually observed that the vehicle tends to slightly "cut" the curves or to experience larger drifts compared to the $\mathcal{H}_2$ case, especially during more abrupt dynamic maneuvers.

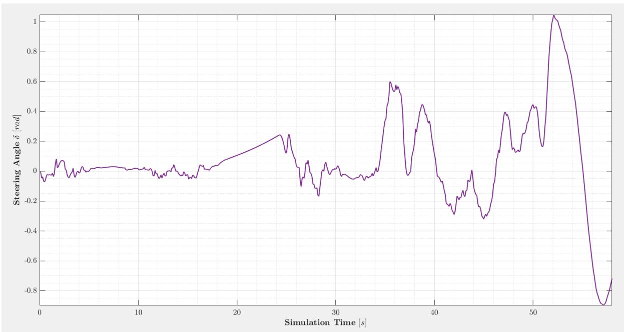

Figure 11: Front Steering angle with $\mathcal{H}_\infty$ controller

As shown in the Figure (11), the $\mathcal{H}_\infty$ controller produces a much smoother and damped control signal, effectively filtering high-frequency components of the disturbance.
Regarding tracking, it eliminates the steady-state error while allowing temporary oscillations and a slower response in exchange for increased robustness.

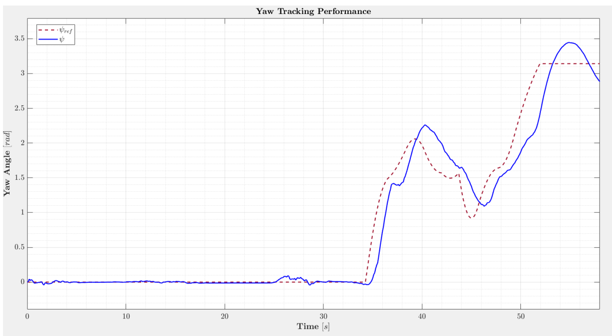

Figure 12: Yaw Angle Tracking with $\mathcal{H}_\infty$ controller

6. Comparison

The comparison highlights the trade-off between precision and robustness. The $\mathcal{H}_2$ controller achieves very small tracking errors but exhibits high-frequency oscillations due to reaction to stochastic disturbances. The $\mathcal{H}_\infty$ controller produces a smoother control signal with fewer high-frequency components, at the cost of larger transient errors.

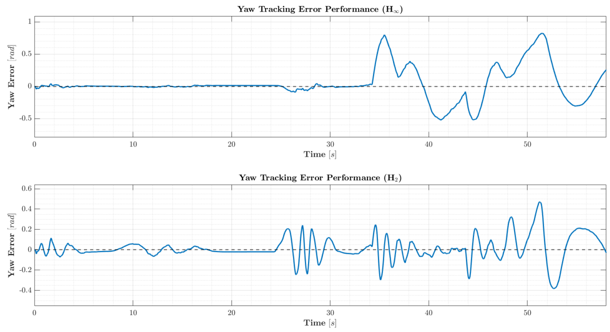

Figure 13: Yaw Angle Tracking Errors

7. 3-D Simulation

For the final validation, the Automated Driving Toolbox was used to recreate an urban scenario within a photorealistic 3D environment. Only the $\mathcal{H}_2$ controller was implemented, as it demonstrated superior tracking accuracy compared to the $\mathcal{H}_\infty$ approach.

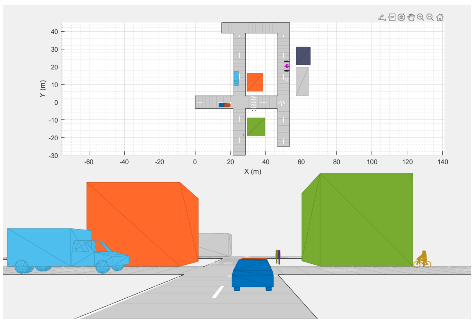

Figure 14: 3D Simulation Scenario at a Road Intersection

Due to the disturbance, the position is not precise.

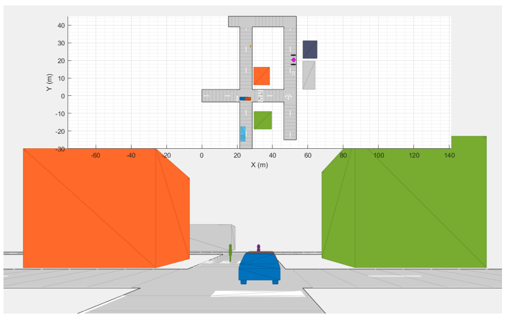

Figure 15: 3D Simulation Scenario at a Pedestrian Crossing

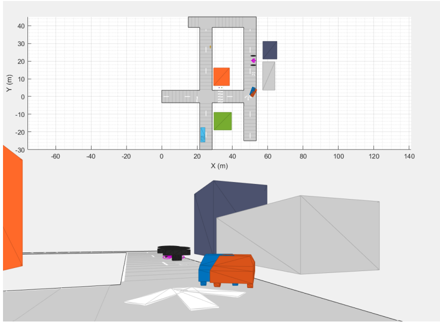

Figure 16: 3D Simulation of the mandatory left turn maneuver

The vehicle performs the left turn accurately, exhibiting only slight yaw deviations.

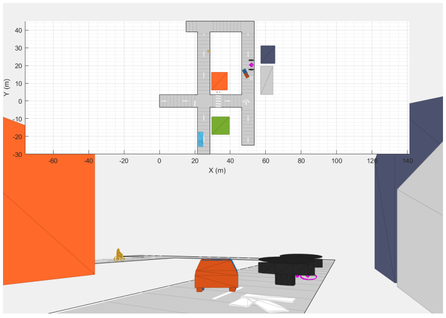

Figure 17: 3D Simulation of the Obstacle avoidance maneuver

The vehicle exhibits nearly zero error while performing the bypass maneuver.

7.1. Position Evolution

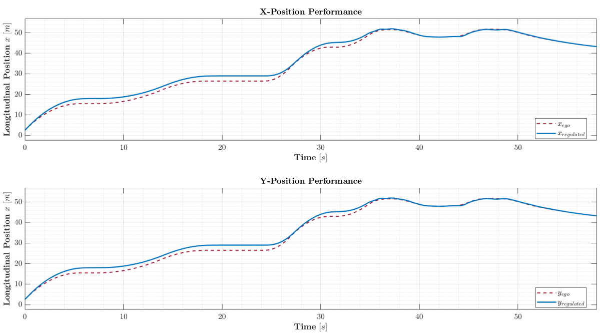

Figure 18: 3D Simulation of the Obstacle avoidance maneuver

The graph shows the perfect overlap between the trajectories of the ideal vehicle (blue) and the controlled vehicle (red). Despite the stochastic disturbance, the $X - Y$ error remains negligible even during the turn and obstacle bypass, validating the high precision of the $\mathcal{H}_2$ control.

8. Conclusions

This report investigated the implementation of robust control strategies for an autonomous vehicle in a complex urban environment. The study successfully modeled the vehicle dynamics using a Linear Parameter-Varying (LPV) framework to account for variations in longitudinal velocity.

This study compared $\mathcal{H}_2$ and $\mathcal{H}_\infty$ robust controllers for autonomous urban navigation using an LPV model.

The simulation results highlight a clear performance trade-off:

- **$\mathcal{H}_2$ Control**: Provides superior tracking accuracy and negligible steady-state error, though it is more sensitive to stochastic noise, resulting in high-frequency steering oscillations.
- **$\mathcal{H}_\infty$ Control**: Offers a smoother, more damped response with better noise rejection at the expense of larger transient errors and slower tracking.

The final 3-D Simulation confirms the $\mathcal{H}_2$ controller as the most effective for scenarios requiring high precision, such as obstacle avoidance and tight turns.